

JIAYU ZHANG

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Education

Columbia University, Ph.D. in Operations Research, Aug. 2024 - **Present**

- Advisor: Tianyi Lin (<https://ieor.columbia.edu/content/tianyi-lin>)

Columbia University, M.S. in Operations Research, Aug. 2024 - Oct. 2025

Shanghai Jiao Tong University, B.S. in Mathematics, Aug. 2020 - Jun. 2024

Publications

Preprints

- Goldfarb, Donald, Lexiao Lai, Tianyi Lin, and Jiayu Zhang. "Non-Convex Self-Concordant Functions: Practical Algorithms and Complexity Analysis." arXiv preprint arXiv:2511.15019 (2025).
- Lei, Mingyu, Jiayu Zhang, and Yinyu Ye. "Blessing of high-order dimensionality: from non-convex to convex optimization for sensor network localization." arXiv preprint arXiv:2308.02278 (2023).

Undergraduate Thesis

- Zhang, Jiayu. "Accelerated Gradient Descent with Long Step-Size." Shanghai Jiao Tong University (2024). Supervisor: Jingwei Liang.

Experiences

Cardinal Operations, Student Researcher, Feb. 2023 - Sep. 2023, Shanghai, China

- Demonstrated that stochastic first-order methods reduce computation time by ~90% for large-scale (>10,000 nodes) sensor network localization problems by benchmarking first-order and semi-definite programming based methods with commercial solvers.
- Developed theories on the geometry of the sensor network localization problem, establishing the foundation for a co-authored research paper.

Teaching

Columbia University, Teaching Assistant

- Fall 2025: IEOR 6613 Optimization I (Ph.D. level), instructor: Vineet Goyal

- Spring 2025: IEOR 4004 Optimization Models and Methods I (graduate level), instructor: Vineet Goyal

Fellowships and Awards

- Fall 2024: Boyle Fellowship, Columbia University

Skills

- Coding: Python (PyTorch, NumPy, SciPy, Pandas), Linux, Latex, Julia, Matlab
- Languages: English, Mandarin